Equipment Booking System

Advanced Java (CSE 456) — Group Project

A web-based system for managing borrowing and returning learning equipment in university settings.

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The Challenge of Manual Equipment Management

Tracking Difficulties

Manual systems make it hard to locate equipment and verify availability in real-time.

Borrowing Management

Tracking who borrowed what, when it was due, and whether it was returned becomes error-prone.

No Centralized System

Disparate records across departments lead to lost equipment and inefficient resource allocation.

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Project Objectives

01

Manage Equipment Inventory

Centralized catalog of all available learning equipment with categories and availability tracking.

02

Enable Equipment Transactions

Streamlined borrowing and returning processes with automatic inventory updates.

03

Track Borrowing History

Complete audit trail of all equipment transactions for accountability and reporting.

04

Admin Dashboard

Comprehensive management interface for overseeing users, equipment, and bookings.

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System Features

User Features

- Secure login and authentication
- Browse and search equipment catalog
- Check equipment availability
- Request and confirm bookings
- Return equipment with status update
- View personal borrowing history

Admin Features

- Full CRUD operations on equipment
- Manage equipment categories
- Create, edit, and deactivate user accounts
- View all booking records
- Monitor system activity
- Generate usage reports

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Technologies Used

Spring Boot

Backend framework providing rapid development, dependency injection, and embedded server capabilities.

Java

Core programming language leveraging object-oriented principles and enterprise features.

MySQL

Relational database management system for persistent data storage and retrieval.

Thymeleaf + Bootstrap

Templating engine and frontend framework for responsive, dynamic web interfaces.

Hibernate / Spring Data JPA

ORM layer simplifying database interactions and reducing boilerplate code.

Apache Maven

Build automation and dependency management tool for project compilation and deployment.

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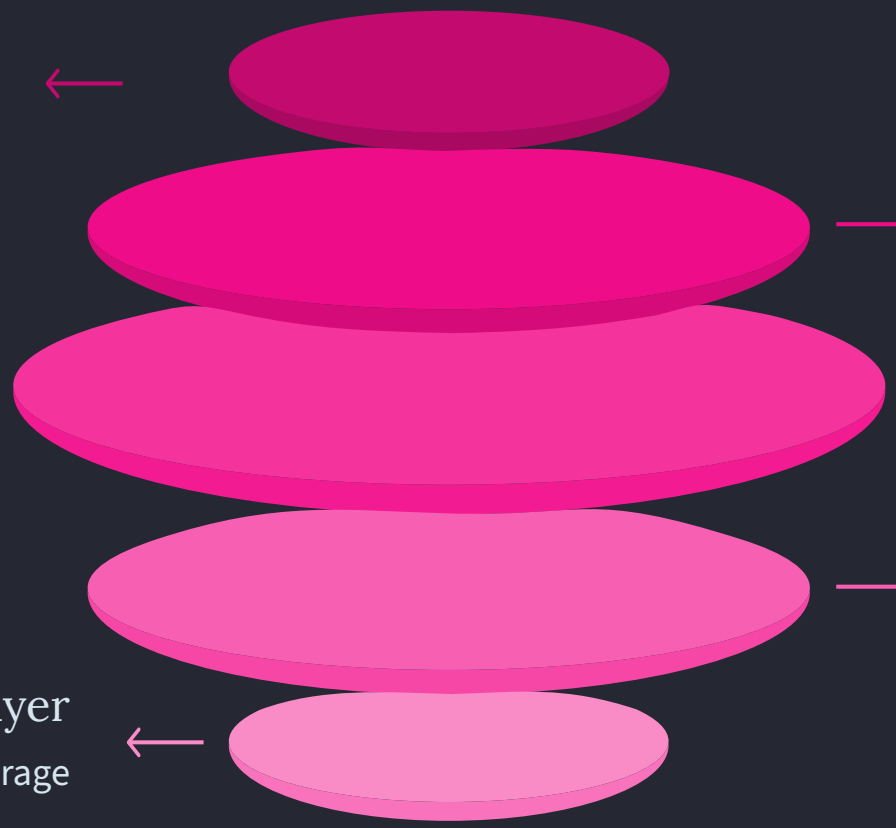
System Architecture

Our system follows a layered architecture pattern, ensuring separation of concerns and maintainable code structure.



Client Layer

User interface and input

Controller Layer

Request handling and routing

Service Layer

Business logic and orchestration

Repository Layer

Data access and mapping

Database Layer

MySQL persistent storage

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Database Design

1

Accounts Table

User credentials, roles, and contact information with unique identifiers.

2

Equipment Table

Equipment details including name, description, quantity, category, and availability status.

3

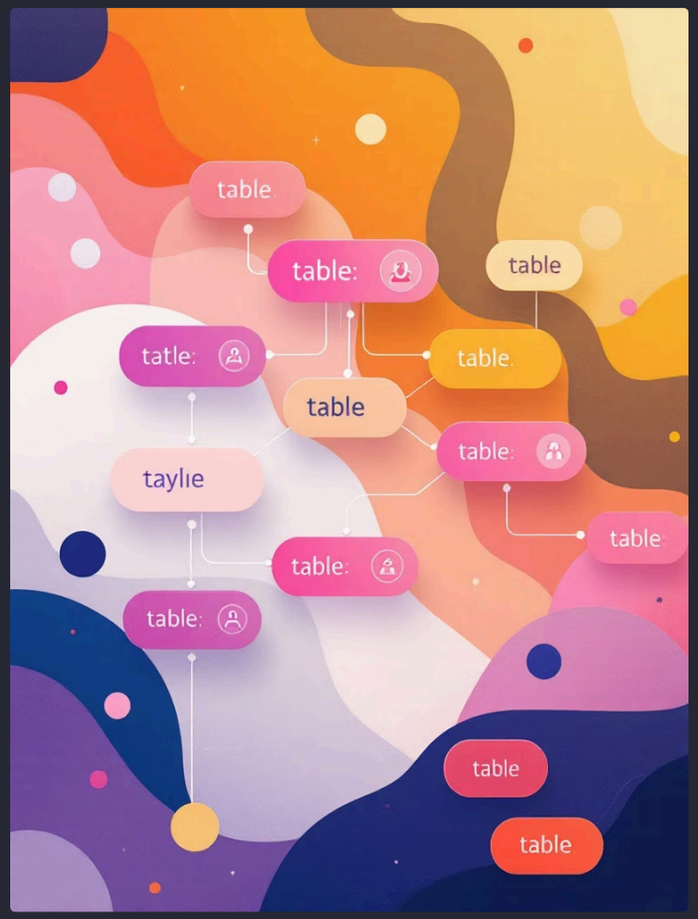
Categories Table

Equipment classification system for organized browsing and searching.

4

Bookings Table

Transaction records linking users to equipment with timestamps and status tracking.



This relational structure efficiently tracks who borrowed what equipment, when, and whether it has been returned.

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Borrowing Workflow

User Login

Authentication validates user credentials and establishes session.

Select Equipment

User browses catalog and chooses desired equipment from available items.

Check Availability

System verifies equipment is in stock and not already booked.

Create Booking

Booking record is created with user ID, equipment ID, and timestamp.

Update Inventory

Available quantity decreases automatically, preventing overbooking.

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Security Features

Session Authentication

Secure HTTP sessions prevent unauthorized access and session hijacking through server-side session management.

Role-Based Access Control

Different permission levels (Admin vs. User) ensure users can only access appropriate features and data.

Input Validation

Server-side validation prevents SQL injection, XSS attacks, and data integrity issues before processing requests.

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Future Improvements

Password Encryption

Implement BCrypt hashing to securely store user passwords with salted hashes.

Email Notifications

Automated alerts for booking confirmations, reminders, and overdue equipment.

Report Generation

Exportable reports on equipment usage, popular items, and department statistics.

Mobile Support

Responsive redesign or dedicated mobile app for booking equipment on-the-go.

Conclusion

This system demonstrates practical application of Spring Boot, Java, and MySQL for enterprise web development, providing universities with an efficient solution for equipment management while showcasing modern software engineering practices.

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